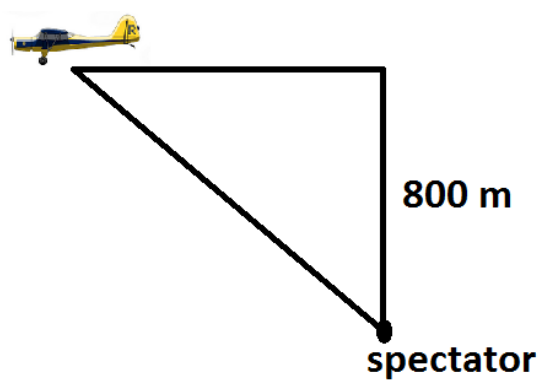
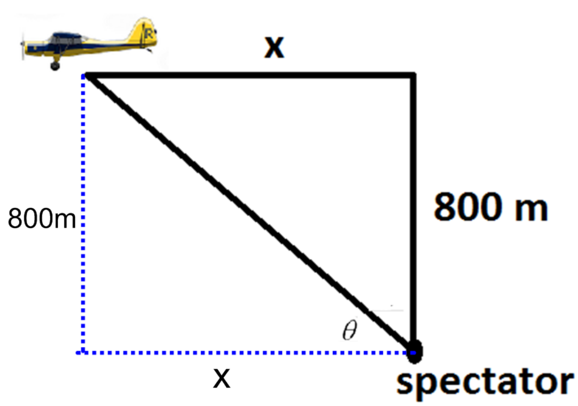
**Problem 3**

# Problem 3

**Problem 1** *A plane flying horizontally at an altitude of 800m and speed of 200 m/sec passes directly over a spectator at an air show. (See figure below.) Find the rate at which the angle of elevation, $\theta$, is changing 4 seconds later. Make sure to label the figure. ($\theta$ = angle between the ground and the line from the spectator to the plane)*




**Explanation.**

$$
\begin{aligned}
\tan\theta &= \frac{800}{x} \\
\sec^2\theta \cdot \frac{d\theta}{dt} &= \frac{-800}{x^2} \cdot \frac{dx}{dt} \text{ (see } x \text{ and } \sec(\theta) \text{ solved below)} \\
2 \cdot \left[\frac{d\theta}{dt}\right]_{t=4} &= \frac{-800}{800^2} \cdot 200 \\
\left[\frac{d\theta}{dt}\right]_{t=4} &= \frac{-800}{800^2} \cdot 200 \cdot \frac{1}{2} = -\frac{1}{8} \text{radians per second}
\end{aligned}
$$

To solve for $x$ when $t = 4$:
Use $d = rt$. Here we have $x = rt$. $x = (200)(4) = 800$ m

To solve for $\sec\theta$ when $t = 4$:
Since the two legs of the triangle are equal, $\theta = \frac{\pi}{4}$
$\cos\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}} \implies \sec\left(\frac{\pi}{4}\right) = \sqrt{2}$